

Security Operations Center

Lab 3: Blue teaming exercises

Lecture goals

The lecture aims to achieve the following goals:

1. Provide basic knowledge on how to use Sentinel
2. Provide practical experience with building, testing and executing detection queries
3. Provide practical experience with building threat hunting queries
4. Have fun and learn by experimenting

Ground rules

1. The exercises are interactive in nature, expect to talk
2. The exercises are structured in a progressive manner and will last 20 minutes each
3. Work with the group to which you are assigned and help each other
4. Nominate one group lead who will participate in the exercise outcome discussions, be prepared to support the team lead during the outcome discussions
5. At least 3 groups will be nominated to present their solution/outcome before each exercise
6. Expect at least one group to be asked to present their solution/outcome for 10 minutes after each exercise
7. There is no failing – this is a safe place where you can experiment and learn. In fact, if you struggle to complete the exercise explain to the class what blocked you and what you think you must do to improve (i.e. what knowledge do you feel you are missing, what extra information could the instructor have given you etc.)
8. Equally, make sure to explain what information helped you fully/partially achieve the goal of the exercise
9. If needed, the solution to each exercise will be given to you

Exercise I

Goal

Build your first detection query using MITRE ATT&CK

Guidance

To build your first Kusto query you must:

- Use the *Logs* blade to inspect logs
- Use the *Sysmon* command stored within Sentinel
- Write a query to detect instances of ATT&CK technique *T1018* being used

Hints

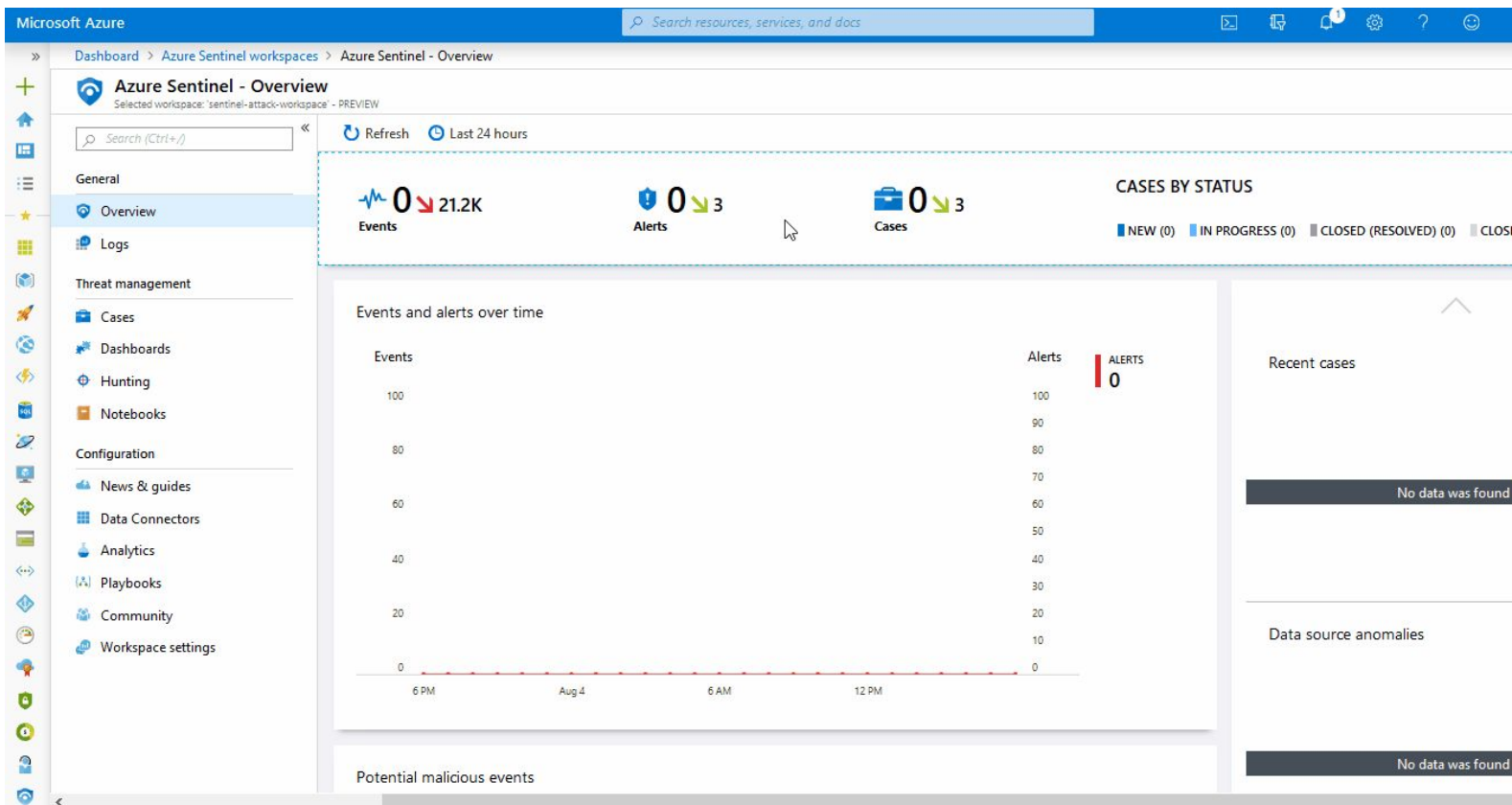
- Try and find instances of *net.exe* commands being issued on virtual machines. Remember that on Sentinel you can see activity being performed on all virtual machines in the lab
- Go to <https://github.com/netever/sentinel-attack/tree/dev/v2/detections> for inspiration

Exercise I

Outcome discussion

Exercise I

Solution discussion



Exercise II

Goal

Build your first detection rule in Sentinel and test it

Guidance

To build your first Kusto detection rule you must:

- See if you can find the *Scheduled rule* page to create detections, if not just use the *Logs* blade to experiment
- Study the ATT&CK technique assigned to your group, as specified in the table on the right
- Log onto your group's virtual machine and execute the technique, then see if it triggered in Sentinel

Hints

- Be mindful that Sentinel is being merged with Defender, the *scheduled rule* page may be hard to find
- Go to <https://github.com/edoardogerosa/sentinel-attack/tree/dev/v2/detections> for inspiration

Group number	ATT&CK Technique
1	T1033
2	T1018
3	T1007
4	T1012
5	T1016
6	T1040
7	T1057
8	T1059
9 + 10	T1087
11 + 12 + 13	T1201

Exercise II

Outcome discussion

Exercise II

Solution discussion

The screenshot displays the Microsoft Sentinel 'Logs' workspace. The left sidebar contains a navigation menu with options: General, Overview, Logs (selected), Guides, Search, Threat management, Content management, Configuration, Workspace manager (Preview), Data connectors, Analytics, Summary rules, Watchlist, Automation, and Settings. The main area features a 'New Query 2*' tab with a search bar and a 'Run' button. Below the search bar, the time range is set to 'Last 24 hours' and the number of results is '1000'. The query editor shows a single query: '1 | Type your query here or click one of the queries to start'. Below the editor, the 'Query history' section lists three queries:

Query	Results	Time	Action
SecurityEvent where * has 'Sysmon'	43550 results	07/05/2026, 09:30	Run
Sysmon	74 results	07/05/2026, 09:21	Run
Sysmon where Computer contains "pc8-"	1000 results	04/05/2026, 18:10	Run

At the bottom left, a small note reads: 'Add or remove favorites by pressing Ctrl+Shift+F'.

Exercise III

Goal

Build a simple threat hunting query using Cyber Threat Intelligence

Guidance

To build your first threat hunting query you must:

- Read the Lazarus group threat intelligence report from F-Secure (found in the lab exercise folder on ILIAS) and identify a malicious command to test
- Identify the right Sysmon event to be targeted during the hunt using the provided Sysmon cheatsheet
- Try to find out how you can store your query in sentinel/defender as a saved hunting query

Hints

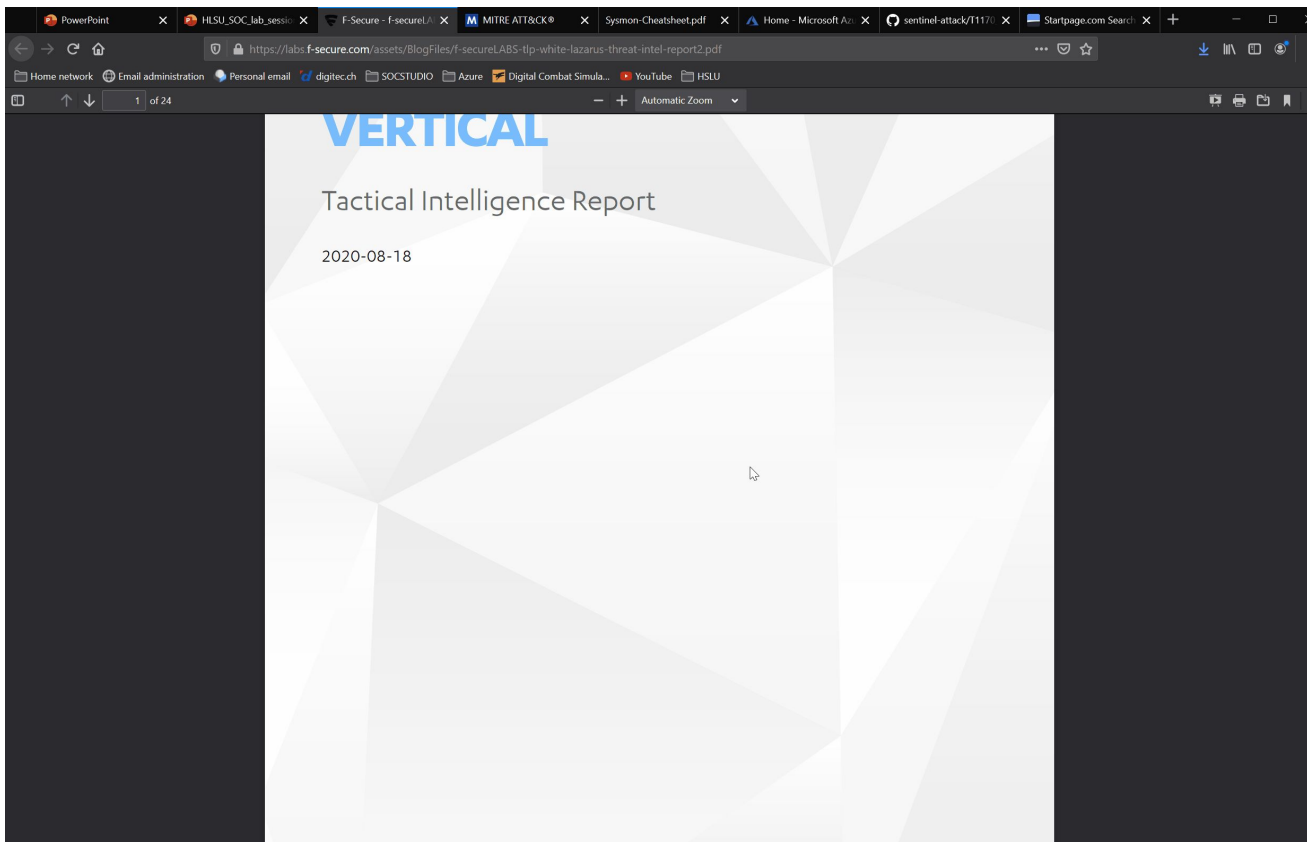
- To save time only read the initial access section of the document, page 5 in particular gives a good example of two indicators of compromise that can be targeted with your hunting query
- Go to https://github.com/edoardogerosa/sentinel-attack/blob/dev/v2/detections/T1170_MSHTA_Process.txt for inspiration

Exercise III

Outcome discussion

Exercise III

Solution discussion



Exercise IV

Goal

Test your threat hunting query on your virtual machine

Guidance

To execute your first threat hunt you must:

- Log onto your group's virtual machine and execute the technique
- Use the *Logs* blade to search for your command
- Test if your saved query identifies the target technique or if the query needs to be refined

Hints

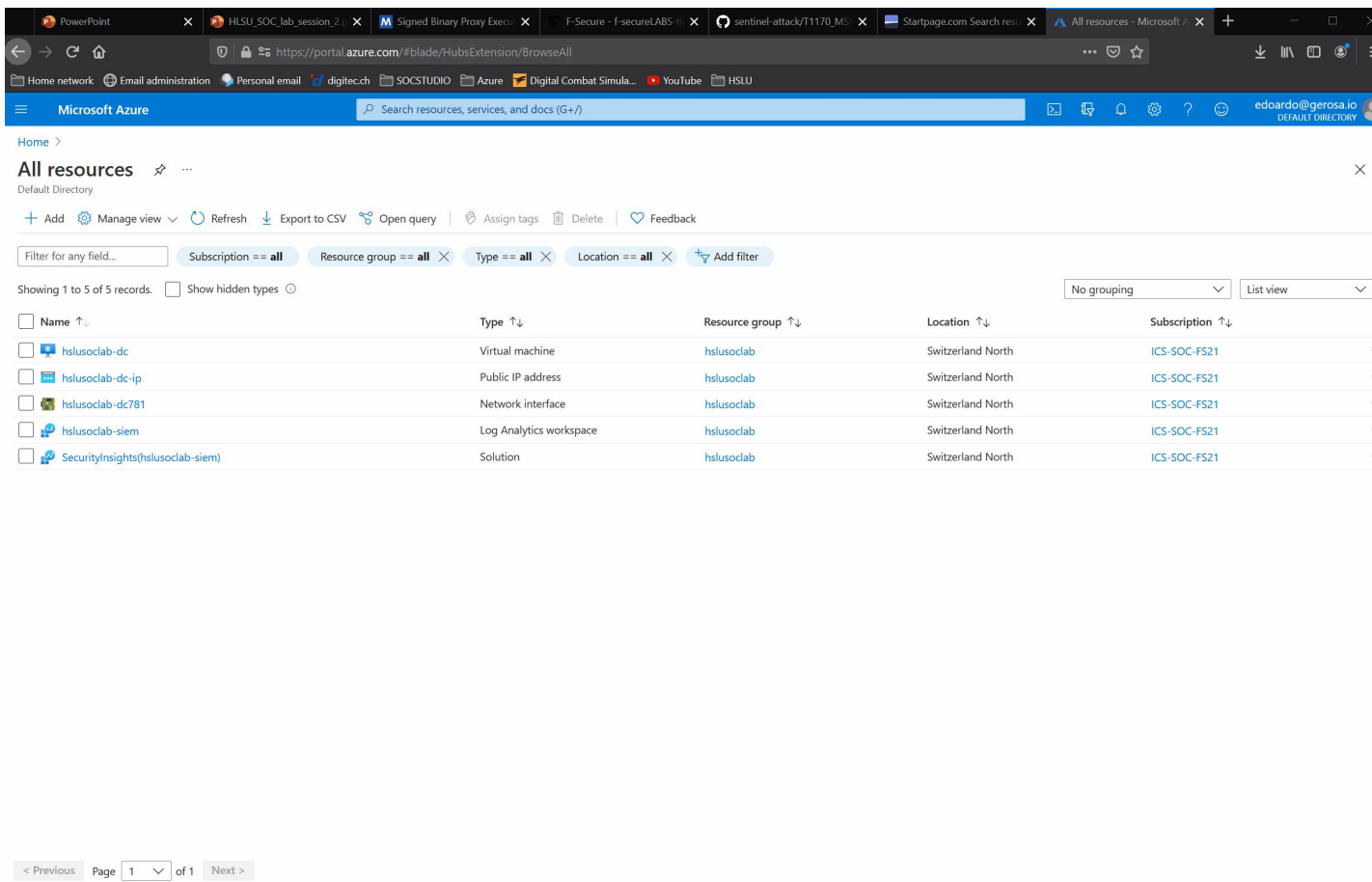
- Use the knowledge gained in exercise II to execute the test on the virtual machine and hunt for the technique in Sentinel

Exercise VI

Outcome discussion

Exercise VI

Solution discussion



The screenshot shows the Microsoft Azure portal interface. The browser address bar displays the URL `https://portal.azure.com/#blade/HubsExtension/BrowseAll`. The page title is "All resources" under the "Default Directory". The user is logged in as "edoardo@gerosa.io".

Below the header, there are filters and a table of resources. The filters are: "Subscription == all", "Resource group == all", "Type == all", and "Location == all". The table shows 5 records, sorted by Name.

Name	Type	Resource group	Location	Subscription
hslusoclab-dc	Virtual machine	hslusoclab	Switzerland North	ICS-SOC-FS21
hslusoclab-dc-ip	Public IP address	hslusoclab	Switzerland North	ICS-SOC-FS21
hslusoclab-dc781	Network interface	hslusoclab	Switzerland North	ICS-SOC-FS21
hslusoclab-siem	Log Analytics workspace	hslusoclab	Switzerland North	ICS-SOC-FS21
SecurityInsights(hslusoclab-siem)	Solution	hslusoclab	Switzerland North	ICS-SOC-FS21

At the bottom, there is a pagination bar showing "Page 1 of 1".

Thank you for your attention